Jinwei **Yao**

Computer Science · Machine Learning System

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" To breathe reality into imagination by crafting intelligent systems."

Education

University of Illinois Urbana-Champaign(UIUC) *Champaign, USA*

(Research-based) Master of Science in Computer Science *2024 - 2026(expected)*

- **Research-oriented Master Program fully funded by UIUC.**
- **GPA: 4.0/4.0.**
- **Research Topics**: System-algorithm co-design for large language models(LLMs).
- **Supervisor:** Prof.Jiaxuan You.
- **Publications:** ICLR'25 Spotlight[C1], ICML'25[C2]
- **Open-source contributions:** SGLang, LM-Evaluation-Harness.
- **Teaching Service:** CS128(Introduction to Computer Science II), CS598-JY2(Topics in LLM Agents).

École Polytechnique Fédérale de Lausanne (EPFL) *Lausanne, Switzerland*

Ph.D. student in Computer Science *2022 - 2023*

- Got the **Ph.D. fellowship** from CS Dept of EPFL.
- Got **5.75/6** in System for Data Management and Data Science, an advanced course for PhD students
- Left EPFL because **no professors here were interested in ML Systems**.

Zhejiang University *Hangzhou, China*

B.Eng. in Electronic Science and Technology *2018 - 2022*

- **Overall GPA:** 3.89/4.00 (87/100).
- **Honors Minor(40/6800+):**: Advanced Class of Engineering Education(ACEE), **Chu Kochen Honors College**.
- **Graduated with Outstanding Thesis Award**.

Research Interest

System-algorithm co-design for Machine Learning(especially LLMs)

Publications

Conference Proceedings

C1. [ICLR'25 Spotlight] DeFT: Decoding with Flash Tree-Attention for Efficient Tree-structured LLM Inference.
Yao, Jinwei*, Chen, Kaiqi*, Zhang, Kexun, You, Jiaxuan, Yuan, Binhang, Wang, Zeke, and Lin, Tao
In International Conference on Learning Representations (ICLR) 2025, abridged in ICLR'24 workshop AGI (Oral).
[Paper] [Code]

C2. [ICML'25] ResearchTown: Simulator of Human Research Community.
Yu, Haofei and Hong, Zhaochen and Cheng, Zirui and Zhu, Kunlun and Xuan, Keyang and **Yao, Jinwei** and Feng, Tao and You, Jiaxuan
In International Conference on Machine Learning(ICML) 2025
[Paper] [Code]

Journals

J1. [HaSS'23] Instruction-Level Power Leakage Evaluation of Soft-Core CPUs on Shared FPGAs.
O. Glamocanin, S. Shrivastava, **J. Yao**, N. Ardo, M. Payer, and M. Stojilovic.
Springer Journal in Hardware and Systems Security, special issue on Multitenant Computing Security Challenges and Solutions, 2023
[Paper]

Experience

[System-algorithm Co-design for LLM]

GRAPE: Graph-based KV Cache Selection for Efficient Long-context LLM Inference. *UIUC, USA*

RESEARCH ASSISTANT AT ULAB, UIUC *Oct. 2024 -*

- **Mentor: Prof.Jiaxuan You(UIUC)**
- **Motivation:** The growing demand for long-context LLMs has led to significant challenges in managing memory and computational overhead due to the expanding KV cache size.
- **Problem:** Existing KV cache selection algorithms for long-context LLMs struggle with inefficiencies due to static or heuristic strategies that lack query adaptability and fail to utilize token importance similarity across layers.
- **Insight:** Token importance often exhibits continuity across consecutive layers, and leveraging this similarity alongside query-aware clustering can significantly improve KV cache efficiency and performance during long-context inference.
- **Idea:** GRAPE is a graph-based KV cache selection algorithm that clusters keys using a Key Neighbor Graph (KNG) during prefill and performs sparse attention during decoding by inheriting and expanding token candidates from prior layers.
- **Technique report:** (As a course project in UIUC CS546 Advanced NLP) Overleaf Link Here.

[System for LLM]

Flash Tree-attention for Efficient Tree-structured LLM Inference. *Westlake University, China*

RESEARCH ASSISTANT AT LINS LAB, WESTLAKE UNIVERSITY *Oct. 2023 - Oct.2024*

- **Mentor: Prof.Tao Lin(Westlake Univ.), Prof.Binhang Yuan(HKUST), Prof.Zeke Wang(Zhejiang Univ.), Prof.Jiaxuan You(UIUC)**
- **Motivation:** Interactions with LLMs are becoming increasingly complex, encompassing tasks like few-shot prompting, multi-step reasoning, and speculative decoding. These tasks often involve multiple generation calls organized in a tree structure with shared token prefixes.
- **Problem:** However, existing inference systems for tree-based applications are inefficient due to improper partitioning of queries and KV cache during attention calculations.
- **Insight:** Tree-based decoding systems for LLMs face inefficiencies caused by redundant KV cache IO for shared prefixes and poor load balancing during attention computations.
- **Goal:** Improve the efficiency of tree-based decoding in LLM inference by reducing KV cache IO overhead and enhancing GPU utilization through optimized attention algorithms.
- **Publications: See C1 in Publications**

[Programming Language & Distributed System]

Performance Optimization for Distributed Agent-based Simulation Engine *EPFL, Switzerland*

RESEARCH INTERN AT DATA LAB. *Apr. 2023 - Jun.2023*

- **Mentor: Prof.Christoph Koch**
- **Motivation:** Large-scale agent-based simulations are popular for data science, while they have seen very little foundational research from core computer science.
- **Challenges:** Scaling agent-based simulations up and out is challenging for its complex data transformation, aggregation, and time series analysis.
- **Goal:** Optimize the performance of agent-based simulations based on previous work-CloudCity.
- **Related Knowledge :** distributed computing, programming language, databases, and computer architecture.
- **Skills: Scala,Java**

[Distributed System]

A Starvation-free Consensus for Decentralized Cross-shard Transaction Commit *EPFL, Switzerland*

RESEARCH INTERN AT DECENTRALIZED AND DISTRIBUTED SYSTEMS LAB(DEDIS). *Oct. 2022 - Feb. 2023*

- **Mentor: Prof.Bryan Ford**
- **Motivation:** Most commit protocols for cross-shard transactions in decentralized systems have starvation problems and are also expensive in communication.
- **Insight:** Starvation is mainly caused by the contentions for the exclusive locks. Transaction atomic commit and consensus have many similarities.
- **Challenge:** 1. How to resolve contention without introducing a global or even an intra-shard leader for ordering. 2. How to reduce communication costs.
- **Solution:** Fair contention resolution to enable starvation-free. Unifying consensus and atomic commit to reducing communication costs.
- **Skills: GoLang**

[System for ML]

Accelerating Graph Neural Network Sampling by Customizing FPGA Subsystem *Zhejiang University, China*

UNDERGRADUATE RESEARCH INTERN AT RC4ML LAB, ZHEJIANG UNIVERSITY *Nov. 2021 - May.2022*

- **Mentor: Prof.Zeke Wang**
- **Motivation:** Graph sampling, as the first step for GNN training, is a time-consuming irregular computation on modern CPU.
- **Goal:** Customize an FPGA-based memory subsystem with High Bandwidth Memory (HBM) for graph sampling acceleration.
- **Hardware level:** High-bandwidth memory(HBM) can be applied to accelerate the sampling.
- **Software level:** 1)Pipelines and loop unrolling can increase parallelism. 2) The optimization of read and write strategies for graph information can squeeze the performance of HBM.
- **Win Outstanding Thesis Award for undergraduates(rank 1/200+).**
- **Skills: C++, Vitis, VHDL**

[FPGA Security]

Power-based Side-channel Disassembly Attacks on Shared FPGA — *EPFL, Switzerland*

SUMMER@EPFL PROGRAM RESEARCH INTERN AT PARALLEL SYSTEMS ARCHITECTURE LABORATORY, EPFL — *Jul. 2021 - Nov. 2021*

- **Mentor:** **Dr. Mirjana Stojilovic**
- **Supported by Summer@EPFL fellowship** (acceptance rate in 2021 was **1.5%**)
- **Motivation:**The voltage fluctuation generated by the softcore CPU deployed on the FPGA may leak side channel information when the instruction is running.
- **Goal:** Based on the voltage information, disassemble CPU instructions.
- **Insight:** The voltage fluctuations can be regarded as time-series signals.
- **Challenges:** 1) How to generate a robust dataset (noise between adjacent instructions needs to be considered)? 2)The critical paths of the same type of instructions (such as Add and Sub) are almost the same. What's worse, the sensor sampling rate on the FPGA is very low, and it is difficult for traditional ML methods to obtain enough information for classification.
- **Solution:** 1) Generating random but balanced training set by designing noise-tolerated templates for different instructions; 2) Using lightweight LSTM+CNN model to do the classification. LSTM is suitable for processing sequence data, and CNN can help LSTM learn more meaningful feature representations.
- **A paper has been accepted by Springer Journal in Hardware and Systems Security(HaSS'23).**
- **Skills: Python, Vivado**

Honors & Awards

INTERNATIONAL

2025	**Spotlight(Top 5.1%)**, ICLR 2025	*Singapore*
2022	**Ph.D. fellowship**, EPFL	*Switzerland*
2021	**Summer@EPFL 21 research fellowship(1.5% applications were awarded)**, EPFL	*Switzerland*

DOMESTIC

2022	**Outstanding Thesis Award(rank 1/200+**, Graduation project for bachelor, Zhejiang University	*China*
2022	**Dean's Outstanding Honor**, Outstanding Graduate of Zhejiang University	*China*
2021	**Winner's Prize** , National College Student Information Security Contest	*China*
2021,2019	**Academic Fellowship and Academic Excellence Award**, Zhejiang University	*China*
2019	**Outstanding Student** , Zhejiang University	*China*

Extracurricular Skills

Language

ENGLISH: TOEFL

- **103 – Listening 25/Reading 29/ Writing 24/ Speaking 25**
- **110(BEST) – Listening 30/Reading 29/ Writing 26/ Speaking 25**

Programming

EXPERIENCED: PYTHON, SCALA, JAVA; FAMILIAR: GOLANG, C, MATLAB, VHDL

- **Framework: Tensorflow, PyTorch, Keras, Vivado, Vitis**